CERN: PHYSICS BEFORE THE PARTICLE NAMES

IB PREPARATION

world → words → model → test → equation

Name ______________________ **Date** __________

THE VISIT QUESTION How can we know what happened in a collision that nobody can see?

1 ACCELERATE: electric fields and energy

IB relation: $W = qV = \Delta E$. Look for radiofrequency cavities and the vacuum beam pipe.

Why must energy be supplied repeatedly?

2 STEER: magnetic fields and circular motion

IB relations: $F = qvB$ and $F = mv^2/r$. Look for bending and focusing magnets.

What does the magnetic field change?

3 COLLIDE: energy and momentum

Track the chosen system, its boundary, total energy and total momentum. Collision energy can become rest energy and kinetic energy.

Why use two head-on beams?

4 DETECT: infer particles from effects

Look for ionisation, deposited energy, curved tracks, light pulses or arrival times.

One signal: ______________

The claim it supports: ______________

THE PHYSICS HINGE: READ ONE EXHIBIT

System	Shared relation	Case-specific physics
Assumption	**Measurement**	**Interpretation**

JUST BEYOND THE IB COURSE: QUESTIONS TO BRING BACK

Quarks and leptons | fermions and bosons | force carriers | decay chains | antimatter | the Higgs field | Feynman diagrams

Which idea did the visit make you want to understand? ______________

BRING HOME THREE THINGS

1. A familiar IB relation operating at an unfamiliar scale ______________
2. A measurement that revealed something invisible ______________
3. A question that lies beyond the present syllabus ______________